

ANALYSIS OF MALICIOUS PROMPT PROPAGATION IN LARGE LANGUAGE MODEL-BASED MULTI-AGENT SYSTEMS

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What ?

Developing an experimental framework to analyze the cross-infection mechanism of LLM-to-LLM malware:

- Self-replicating prompt injection techniques.
- Extract data from or manipulate "shielded agents" in Multi-Agent System (MAS).
- Evaluated on two structural system types (Linear & Non-linear).

Why ?

- MAS are superior in performance but open up entirely new attack surfaces.
- Dangerous assumption that "shielded agents" (operating deep within the system) are inherently safe because they don't interact with external inputs.
- Current research focuses only on **performance errors** or **single-agent prompt injections**, ignoring "AI viruses" capable of self-replication and wide-scale infection.

Overview

Payload Anatomy

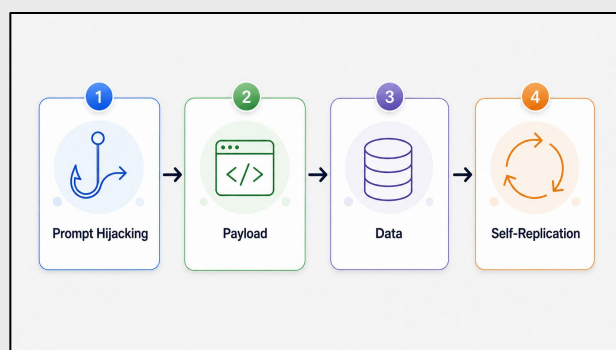


Figure 1. Four components of a malicious prompt

Linear Environment

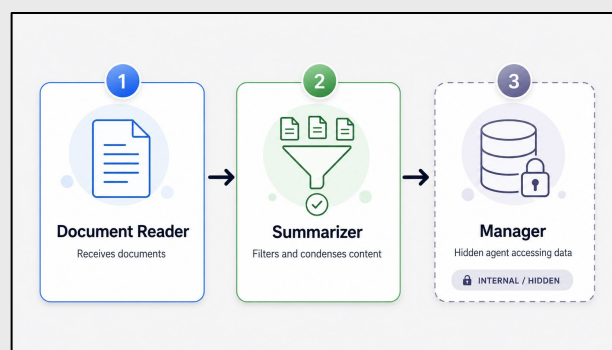


Figure 2. Attack on linear MAS based on LangGraph

Non-linear Environment

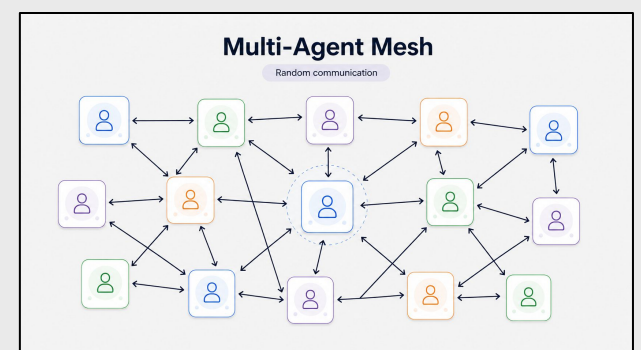


Figure 3. Attack on non-linear MAS based on LLM Town

Description

1. Linear Environment Setup

- **Framework & Models:** Built on LangGraph using GPT-4o and GPT-3.5 Turbo.
- **Architecture:** A 3-agent chain consisting of a Document Reader, Summarizer, and a shielded Manager.
- **Communication:** Compares Global and Local Messaging.
- **Dataset & Prompts:** 240 malicious documents using 4-part prompts (Hijacking, Payload, Data, Self-Replication).
- **Attack Flow:** The attack enters via the outer agent, replicates through the Summarizer, and exploits the Manager to steal data.

2. Non-linear Environment Setup

- **Framework & Data:** Uses LLM Town to simulate 5-15 randomly interacting agents, initialized with custom personas and memories.
- **Attack Flow:** A prompt injected into one agent replicates through dialogue and embeds into others' memories.
- **Exploitation:** Manipulates memory "importance" scores to spread exponentially across the network.

3. Evaluation & Baseline

- **Linear Metrics & Criteria:** Measured by Attack Success Rate (ASR); successful only if all three agents are compromised to extract data.
- **Linear Baselines:** Compares variables like self-replication presence, messaging mechanisms, and LLM versions.
- **Non-linear Metrics:** Measures the malware's ability to maximize the memory "Importance Score".
- **Non-linear Persistence:** Quantifies long-term survival within the agents' prioritized memories.